Analyzing the “Mozart Effect” in Epilepsy Using a Brian2 Spiking Neural Network Model

by

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ABSTRACT

The “Mozart Effect” has been studied for decades, yet there is no unifying explanation for this phenomenon. A novel Brian2-based spiking neural network model was developed to simulate epileptic brain activity during IED (interictal epileptiform discharges). Mozart’s K448 was applied to the model through a cochlea-inspired encoding method. The results revealed variable changes in synaptic weights and spike trains. However, raster plots of the model revealed similarities to real IED data. This research demonstrates the potential of Brian2-based SNNs for computational modeling of the epileptic brain activity.

1 Introduction

The "Mozart Effect" describes an interesting phenomenon where listening to Mozart's K448 reduces the occurrence of seizures in epilepsy patients (Quon et al., 2021). It has been documented to specifically reduce occurrences of IED and increase occurrences of delta- and theta-band activity in the frontal lobe (Hughes et al., 1998). However, there is currently no unifying theory that explains the mechanism of this effect. The non-invasive nature of EEG limits the analysis of brain dynamics. To address this limitation, this study uses the Brian2 simulator to construct a simplified cortical network driven by frontal EEG recordings exhibiting IED and analyze changes that occur as music-derived input is applied. Specifically, the synaptic weights and the van Rossum distance will be compared. This approach may provide deeper insight into the "Mozart Effect" by demonstrating changes in synaptic connectivity and firing patterns through computational means.

2 Background

2.1 Spiking Neural Network

2.1.1 Spiking Neural Network

A Spiking Neural Network (SNN), unlike an Artificial Neural Network, uses neurons that mimic biological neurons, processing both the intensity and timing of input. SNN neurons store voltage, which increases when a synaptic input is received and emits a spike when the voltage reaches the threshold. Input spikes decay over time, enabling the encoding of temporal information. Disadvantages of SNNs include lower performance compared to ANNs and challenges in applying backpropagation for supervised learning (Yamazaki et al., 2022). However, SNNs outperform ANNs in some tasks like brain disease classification, and there have been attempts to introduce surrogate gradients for SNNs. For example, SNNs have been successful in classifying the MNIST handwritten digit dataset, achieving 95% accuracy comparable to ANNs (Diehl & Cook, 2015). SNN models have also been used to model biological observations through Alzheimer's Disease modeling (Capecci et al., 2016) and identified specific EEG channels as biomarkers for Epilepsy (Kasabov, 2023).

2.1.2 Leaky Integrate-and-Fire Model and Spike-Timing Dependent Plasticity

One SNN model is the Leaky Integrate-and-Fire (LIF) model. The LIF model reproduces the leaking of positive ions through the exponential decay of voltage, fires a single spike when the threshold is reached, and resets its voltage to the reset potential. Due to its balance between biological plausibility and computational efficiency, it is the most commonly used neuron model for SNN. The differential equation for an LIF neuron is as follows (Gerstner, 2014):

$$\tau_m \frac{du}{dt} = -\ (u(t)\ -\ u_{rest})\ +\ R\,I(t)$$
$$\tau_m = R\,C$$

Equation 1: LIF Neuron Model

The general solution yields the following exponential equation:

$$u(t)\ -\ u_{rest}\ =\ \Delta u\ exp(-\ \frac{t - t_0}{\tau_m})$$

Equation 2: General Solution of LIF Neuron Model Equation

SNNs can utilize the Spike-Timing Dependent Plasticity (STDP) mechanism for the learning mechanism. This mechanism mimics the time-dependent signaling of biological neurons by increasing synaptic weights when an output spike occurs after an input spike. For example, if Neuron X fires a spike that Neuron Y receives, and Neuron Y fires shortly after, the synaptic weight between Neuron A and Neuron B will increase. The opposite effect will take place if Neuron Y fires before Neuron X. The equation for STDP learning is as follows:

$$\Delta w\ \ =\ \ \pm A\ exp(\pm\frac{t_{pre} - t_{post}}{\tau})$$

Equation 3: STDP Learning Mechanism

2.1.3 Brian2 Simulator

Brian2 is an intuitive Python-based simulator for SNN (Goodman & Brette, 2008). It is a reliable tool that has high flexibility. Parameters such as learning rate, refractory period, and number of neurons can be easily edited, and different data can be easily plotted.

2.2 Epilepsy

2.2.1 Epilepsy

Epilepsy is a neurological disorder characterized by recurrent seizures due to neuronal hypersynchronization. Affecting over 65 million people worldwide, it is one of the most burdensome brain disorders (Devinsky et al., 2018). A non-epileptic brain can undergo epileptogenesis through brain insult, which includes stroke, traumatic brain injury, infectious diseases, etc. Epilepsy diagnosis is a multi-step process that is often overlooked, leading to misdiagnosis. They must first assess medical history for any epilepsy-inducing events, perform physical examination, and routinely check EEG signals for epileptiform discharges. Interictal Epileptiform Discharges (IEDs) are an important biomarker for seizures that occur during interictal, or in-between periods of seizures.

The primary treatment for epilepsy is anti-seizure drugs, which are effective in ⅔ of patients. Some of the remaining one-third of patients who are resistant to drugs undergo a surgical resection, which cures epilepsy. However, the criteria for surgery are high, and most refractory epilepsy patients can undergo surgical treatment (Devinsky et al., 2018). To those patient groups, alleviating occurrences of seizures is the best course of action. This can be done through neurostimulation devices, dietary therapies, and music.

2.2.2 Auditory Processing

Humans have enjoyed music for tens of thousands of years, and it is not surprising that music greatly influences brain function. Transduction of auditory signals begins in the cochlea, where hairs of the basilar membrane convert auditory signals to firing of the auditory nerve based on pitch and volume (NIDCD, 2022). Then, the signals are processed in the temporal lobe through the primary auditory cortex and associated areas. The Predictive Coding of Music (PCM) theory states that music perception results from the brain trying to minimize prediction error by changing internal predictions. Specifically, the melody, harmony, and rhythm are the main components that assist this, where Mismatch Negativity (MMN) responses resulting from a mismatch engage the brain in music (Vuust et al., 2022).

2.2.3 Music and Epilepsy

In one of the most recent studies with refractory epilepsy patients, patients were exposed to 15 seconds to 90 seconds of Mozart's Sonata for Two Pianos in D major, also known as K448. After 30 seconds of exposure, the IED rate decreased by 66.5%. Other music, like Wagner's Lohengrin Prelude to Act I, showed non-significant results (Quon et al., 2021). This suggests that characteristics of K448, including melody, harmony, and rhythm, may influence brain activity. K448 has been shown to affect the frontal lobe by increasing its theta and delta band power (Quon et al., 2021; Hughes et al., 1998). However, deeper insight is needed to explain how this changes the biology of the brain.

3 Experimental Setup

3.1 Process Flow Chart

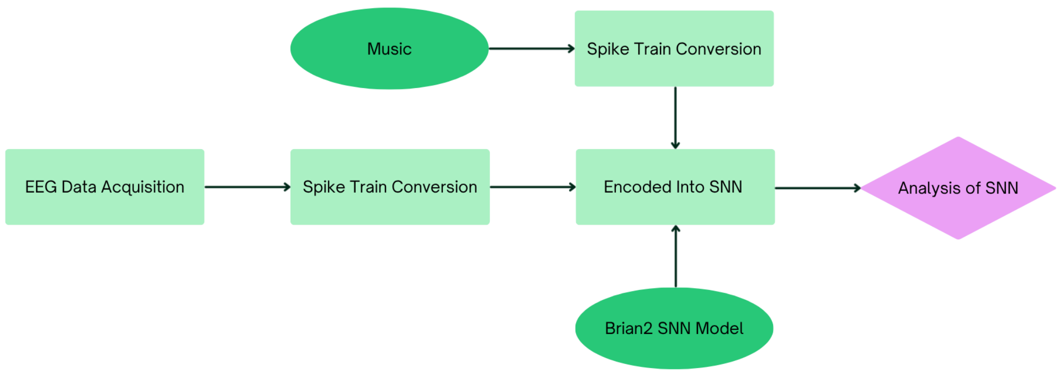

Figure 1: Process Flow Chart

3.2 EEG Processing

3.2.1 Epilepsy IED EEG Data

EEG data of four pediatric epilepsy patients were gathered from the Central University of Kashmir and the Institute of Mental Health and Neurosciences (OK et al., 2021). The data contained segmented EEG data during IED in epilepsy patients with bipolar montage with the following channels: 'FP2-F4', 'F4-C4', 'C4-P4', 'P4-O2', 'FP1-F3', 'F3-C3', 'C3-P3', 'P3-O1', 'FP2-F8', 'F8-T4', 'T4-T6', 'T6-O2', 'FP1-F7', 'F7-T3', 'T3-T5', 'T5-O1', 'FZ-CZ', and 'CZ-PZ'. Bipolar montage measures voltage differences between adjacent channels. The data were preprocessed with a low-pass filter at 70 Hz and a high-pass filter at 1 Hz, were sampled at 256 Hz, and contained 18 EEG channels placed according to the 10-20 system. Files of ten-second segmented data epochs that contained IED were prepared for each patient. Epoch files were concatenated and analyzed through the MNE-Python package. As an example, the Power Spectral Density (PSD) of one patient's recording is shown below.

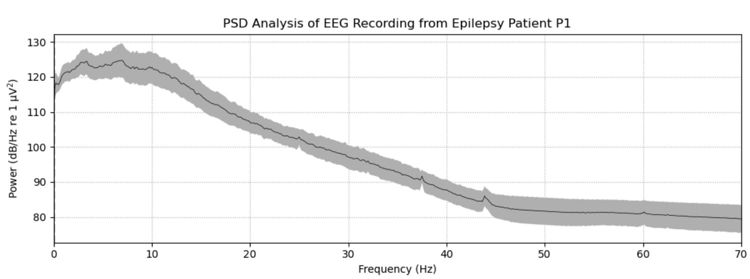

Figure 2: PSD Analysis of EEG Recordings From Epilepsy Patient P1 Exhibiting IED

3.2.2 EEG Spike Train Conversion

The Address Event Representation (AER) method was employed to convert EEG data into a spike train. AER is an encoding method that generates spikes only when significant differences are observed. It conserves both spatial and temporal data, and is commonly used for neuromorphic chips. The threshold value, which determines whether the neuron fires or not through the equation u(t) - u(t-1) > threshold, was chosen to be seven-tenths of the standard deviation of the EEG values of the channel. Although past studies have used fixed thresholds, due to the uniqueness of the bipolar montage, this value was found to be the most suitable threshold. Spikes were divided into positive and negative spikes.

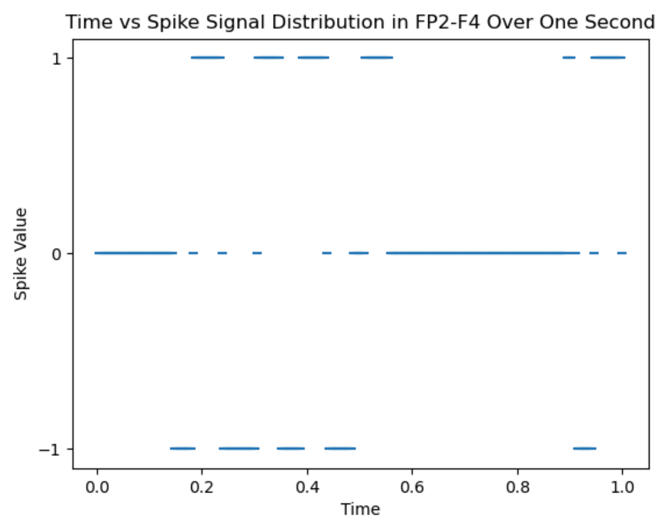

Figure 3: Spike Signal Distribution in FP2-F4 Over One Second

3.3 Brian2 Simulator SNN Model

3.3.1 Model Structure

No standardized Brian2-based SNN framework exists for modeling the "Mozart Effect" on epilepsy. The simplified cortical network was created by closely modeling existing experimental data on neuronal dynamics. The model is a two-layer network combination of a 10 x 10 excitatory neuron grid and a 5 x 5 inhibitory neuron grid. The model size was constrained to reduce computational cost. The specific ratio follows the 8:2 excitatory to inhibitory neuron ratio in the mammalian cortex (Rodarie et al., 2022). The neurons differ in how they shift the postsynaptic voltage; excitatory neurons produce EPSP, and inhibitory neurons produce IPSP. Neurons employed the LIF model and followed the STDP learning rule. Neurons were initialized with weights ranging from 0 to 1, a voltage of 0, and a tau constant of 20. Synaptic connectivity was probabilistic and spatially dependent, where connection probability decayed as the Euclidean distance increased.

$$p = exp(-\sqrt{(x_{pre} - x_{post})^2 + (y_{pre} - y_{post})^2 + (z_{pre} - z_{post})^2}/2)$$

Equation 4: Euclidean Distance-Based Probability

An example figure graph plotting the source and target neuron is shown. Neurons closer together are more densely connected, while neurons far apart have sparser connections.

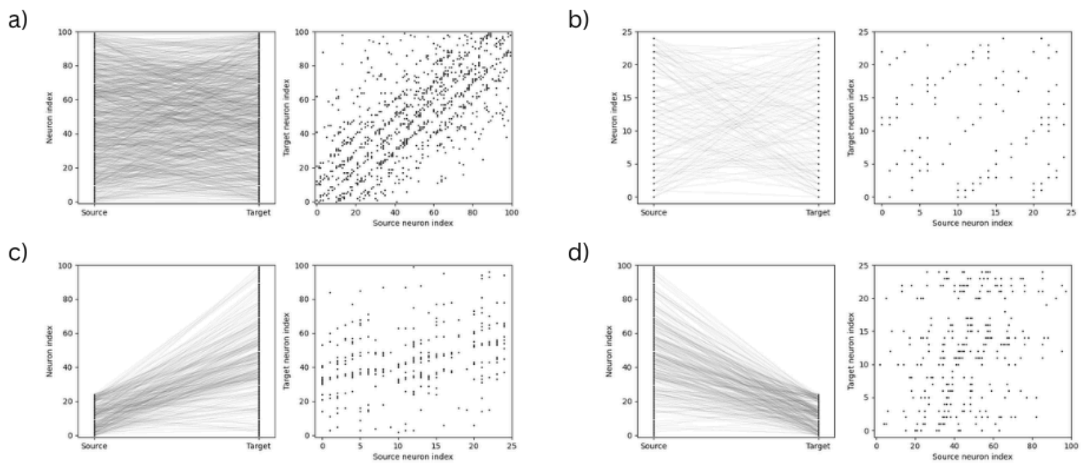

Figure 4: Connection Between a) Excitatory Source and Excitatory Target Neurons, b) Inhibitory Source and Inhibitory Target Neurons, c) Inhibitory Source and Excitatory Target Neurons, and d) Excitatory Source and Inhibitory Target Neurons

3.3.2 Channel Input

The IED EEG-derived spike train of the frontal lobe EEG was used as input to neurons based on the spatial distribution of the EEG channel according to the 10-20 system. The network received 3 minutes of input data, which resulted in 256 Hz X 3 minutes X 60 seconds = 46080 samples. The channel position was based on the 10-20 topology. Positive spikes targeted excitatory neurons, and negative spikes targeted inhibitory neurons. Channels were positioned on the perimeter of the SNN model, with certain channels positioned inside. Z-score normalization was applied. Channels represented were 'FP2-F4', 'F4-C4', 'FP1-F3', 'F3-C3', 'FP2-F8', 'F8-T4', 'FP1-F7', 'F7-T3', and 'FZ-CZ'. Indices of the excitatory neurons were 67, 38, 62, 31, 68, 49, 61, 40, and 35. Indices of the inhibitory neurons were 18, 3, 16, 1, 19, 9, 15, 5, and 2.

3.4 Music Processing

3.4.1 Musical Data

The MIDI file version of K448 was gathered from a MuseScore score (Terushiis, n.d.). It was then processed through Brian2Hears (Fontaine et al., 2011), a computational cochlear model that encodes frequency into channel inputs. Audio values were clipped to be non-negative, raised to the third root, and amplified by a factor of three. The audio was resampled from 44.1 kHz to 10.0 kHz to match Brian2's capacity. A spike was generated every time the signal passed the threshold value, which was chosen based on the patient's average EEG channel firing rate. The spike train was applied on a total of 4 channels, a pair on each lobe used to represent low and high frequency, placed on the left and right temporal lobes for binaural representation. Finally, this spike train played for 30 seconds during the last 150 to 180 seconds of the EEG IED model data.

3.4.2 Control Data

Control data was defined to be the randomized firing of specified channels with a firing rate equal to the average firing rate of music-applied condition channels. The control input was created using the Poisson distribution.

3.5 Analysis of SNN Model

3.5.1 Raster Plot

Raster plots are used in neuroscience to map the timing of spikes in a population of neurons. Although raster plots lack mathematical rigor, they reveal neuronal firing patterns that can provide insight into the general behaviour of the population. During IED, high synchronization is expected to be observed from raster plots. After music is applied, synchrony is hypothesized to decrease.

3.5.2 Synaptic Weight

In SNN, weights can reveal information about the synaptic connectivity of the neuron population. Due to the high amount of neuronal firing occurring during IED and its hyper synchronicity, synaptic weight is expected to be higher than baseline after IED data has been passed into the network. For each patient, the mean synaptic weight across 5 trials was calculated separately for the IED model and the music-applied model. For each patient, values from five trials were used. A two-tailed paired t-test was conducted on each patient ($n = 5$).

3.5.3 Van Rossum Distance

The van Rossum distance is a mathematical method that calculates the similarity of a pair of spike trains (Rossum, 2001). It replaces each spike in the spike trains with an exponential function that decays over time. The difference between the resulting filtered spike train is then squared and integrated, yielding a measure of distance. Comparing the mean van Rossum distance of neurons between the last 30 seconds of the IED model spike train and the spike train during the 30 seconds of application of music can reveal how music shifts spiking dynamics. The van Rossum distance was calculated using the Elephant Python package with a tau constant of 100. This tau value was chosen to compare spike trains based on overall firing rate with less emphasis on strict temporal dynamics. For each patient, the average distance of the music-applied model from the IED model was compared with the average distance of the control model from the IED model. For each patient, distance values from five trials (n = 5) were compared using a two-tailed paired t-test.

4 Data & Results

4.1 Raster Plot of IED Model

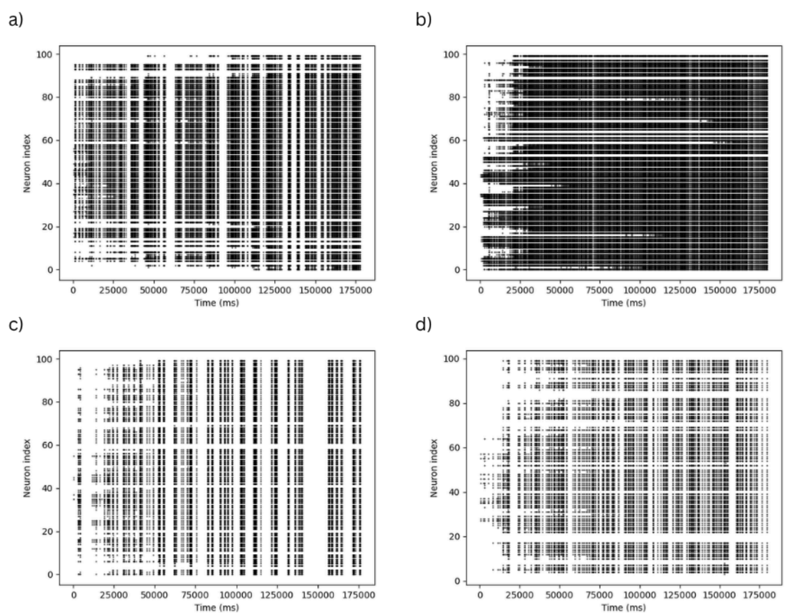

Figure 5: Raster Plot of IED Model of a) Patient One, b) Patient Two, c) Patient Three, d) Patient Four

4.2 Raster Plot of Conditioned Model

4.2.1 Music-Applied Group

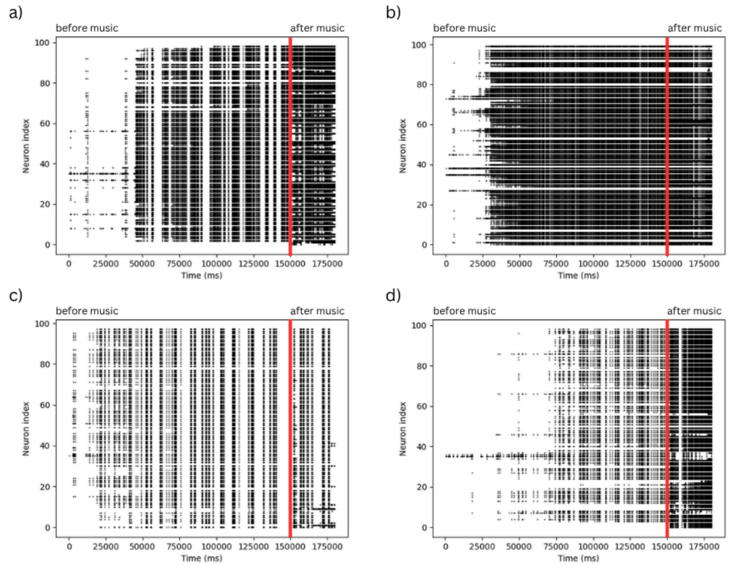

Figure 6: Raster Plot Segment of Music-Applied Model of a) Patient One, b) Patient Two, c) Patient Three, d) Patient Four. Before and After Music Application Distinguished by Red Line.

4.2.2 Control Model

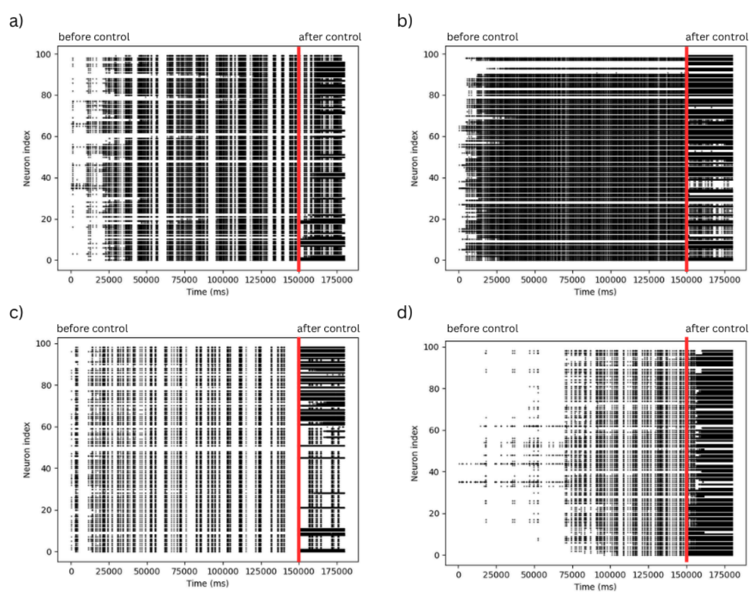

Figure 7: Raster Plot Segment of Control Model of a) Patient One, b) Patient Two, c) Patient Three, d) Patient Four. Before and After Control Input Distinguished by Red Line.

4.3 Synaptic Weight

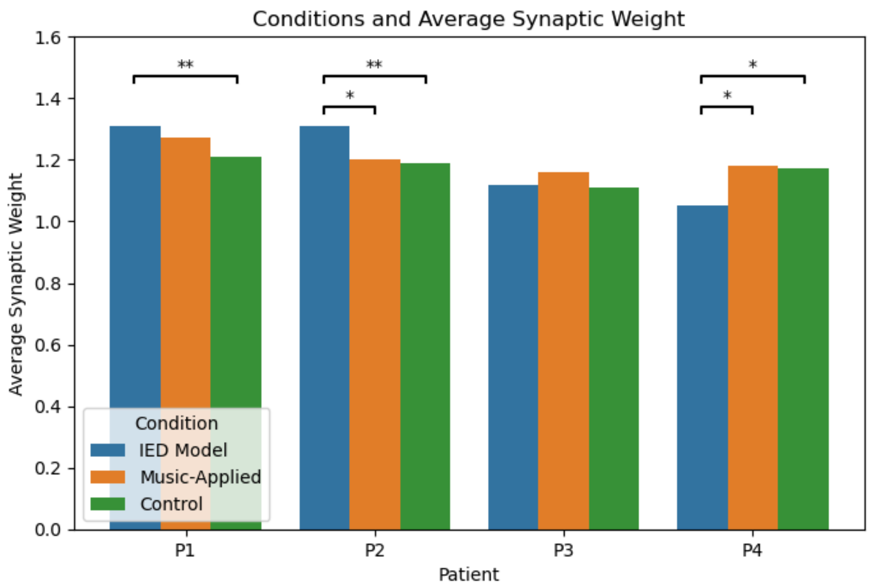

Figure 9: Comparison of Weights Between Conditions. The Significance Bar of "*" Represents $p < 0.1$, While "**" Represents $p < 0.05$. An Empty Significance Bar Represents No Significance.

4.4 Van Rossum Distance

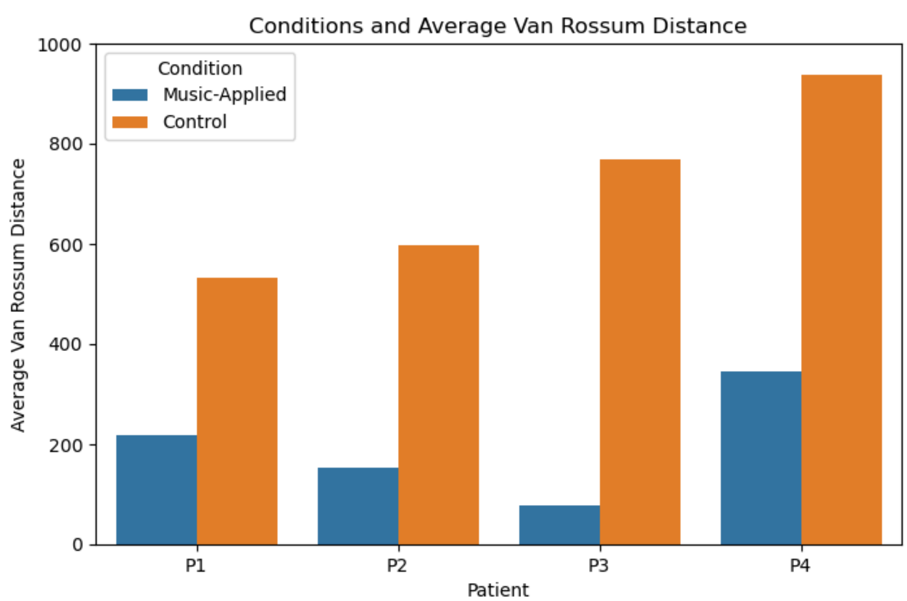

Figure 10: Comparison of van Rossum Distance Between IED Model and Music-Applied Model. The Significance Bar of "*" Represents $p < 0.1$, While "**" Represents $p < 0.05$. An Empty Significance Bar Represents No Significance.

5 Analysis & Discussion

5.1 Raster Plot

5.1.1 IED Model

The IED model displays uncoordinated activity during its initial stage, but exhibits synchronization as more EEG channel inputs are added. The resulting hypersynchronization is a marker of IED, which suggests that the SNN model was able to map certain features of IED.

5.1.2 Conditioned Model

Significant changes occur in neuronal firing patterns for conditioned models, resulting in denser firing patterns. Unlike experimental models, the computational model failed to show that K448 can reduce IED signatures. Furthermore, K448 exhibited similar results to the control condition, supporting a finding by Quon stating that pure frequency encoding fails to capture the complexity of K448.

5.2 Synaptic Weight

The weight changes showed varied results between patients. In patients 1, 2, and 4, the average synaptic weights changed significantly for both music and control conditions, while patient 3 displayed no significant change. Interestingly, patient 4 showed a significant increase in synaptic weight during the conditions. This inconclusive result highlights the strong influence of probability in determining neural connections.

5.3 Van Rossum Distance

The average van Rossum distance between the baseline vs music-applied group and the baseline vs control group had non-significant differences. Converse to expectations, the control group had a larger van Rossum distance value. Surprisingly, the van Rossum distance was highly varied between each trial, ranging from 71 to 1200. Similar to synaptic weight, the van Rossum distance identifies the probability-based synaptic connection as a limitation of the model.

6 Conclusions

6.1 Summary

The study attempted to computationally explain the "Mozart Effect" through the Brian2-based SNN model. Epilepsy patient EEG data exhibiting IED was converted into spike trains used to stimulate the SNN model. Then, music was encoded into a spike train by distinguishing between low and high frequencies.

The raster plot of the IED model reproduced characteristics of IED, such as synchronicity and hyperactivity. This suggests that the SNN model was able to capture certain characteristics of the EEG input, highlighting the potential of the SNN model to model aspects of brain dynamics.

As the conditional inputs were applied to this model, both K448 and the control affected the synaptic weight with variable results. In analyzing the synchrony, the van Rossum distance also revealed highly probability-dependent differences in the overall firing rate of different conditions. Thus, the model failed to capture important brain metrics to explain IED reduction.

Nonetheless, this project is significant because 1) it provided computational evidence consistent with previous findings that pure frequency information alone is insufficient to explain the “Mozart Effect”, 2) it utilized the potential of the Brian2 simulator as an SNN model of the epilepsy brain, 3) applied computational metrics such as the van Rossum distance as a metric, and 4) identified limitations of the model for future references.

6.2 Limitations

Many limitations were identified in this study, as there were no past studies utilizing the Brian2 simulator for simulating epilepsy. First, the EEG data were in an unconventional bipolar montage format, which does not reflect the specific voltage at a channel. This may have affected how IED was represented in the SNN model. Second, the SNN model was highly simplified due to computational limitations and a lack of established references, which may conceal the full effect captured by this model. Third, the cochlea-inspired encoding represented frequency-based encoding and did not capture other musical features such as harmony and rhythm. Fourth, a small number of trials for each patient and the probabilistic nature of the system have led to little to no significance in results, as some EEG/cochlear channels lacked synaptic connections. Future works will need to address these limitations to create a feasible model for the Brian2 simulator of SNN.

6.3 Data Availability Statement

Code uploaded to GitHub: https://github.com/Ilovemanim/Capstone-Project-2026.

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